9.1 Explicitly Unsupported Capabilities

Description

This section defines the capabilities that the system explicitly does not support. Clearly stating unsupported features helps avoid misunderstandings, limits system expectations, and ensures the project remains within its defined scope.

Unsupported Capabilities

- The system shall not support smartphone applications or graphical user interfaces.
- The system shall not require internet access on the farmer’s device.
- The system shall not support languages other than Amharic.
- The system shall not provide fully automated decision-making that replaces human agricultural experts.
- The system shall not execute financial transactions such as buying or selling crops.
- The system shall not provide legal, medical, or non-agricultural advisory services.
- The system shall not store or process biometric identification of callers.
- The system shall not guarantee perfect accuracy for predictions or recommendations.

Rationale

These limitations are necessary to:

- Keep the system accessible through basic phones
- Reduce system complexity
- Maintain safety and reliability
- Align with available data and project timeline

Dependencies

- Project scope definition
- Academic project constraints

9.2 Out-of-Scope Behaviors

Description

This section describes system behaviors and features that are intentionally excluded from the current project scope. These behaviors may be valuable in the future but are not addressed in this graduation project due to time, resource, and scope limitations.

Out-of-Scope Behaviors

- Real-time human operator intervention during calls
- Support for additional local languages or dialects beyond Amharic
- Mobile or web-based farmer dashboards
- Integration with external payment or mobile money systems
- Automatic control of farming equipment or IoT devices
- Real-time satellite or drone data integration
- Continuous online learning using live user data
- Personalized recommendations based on long-term historical profiling

Reason for Exclusion

These behaviors are excluded because:

- They require additional infrastructure and datasets
- They increase system complexity beyond academic scope
- They demand longer development and testing time

- They introduce higher operational and ethical risks

Future Considerations

Although out of scope, these behaviors may be considered in future system extensions if:

- More resources become available
- Additional datasets are obtained
- Regulatory and ethical approvals are addressed

Dependencies

- Project timeline and academic constraints
- Available datasets and infrastructure

10.1 Speech and Text Data

Description

The system shall store and process speech and text data generated during farmer interactions with the AI voice assistant. This data supports speech recognition, recommendation processing, response generation, system improvement, and troubleshooting. All data handling shall respect privacy and security requirements.

Data Types

- Raw voice call audio (caller speech)
- Processed audio segments used for ASR
- Transcribed Amharic text from ASR
- Generated response text before text-to-speech conversion
- Conversation context data for multi-turn interactions

Data Sources

- Incoming phone calls from farmers
- AI-generated responses from the system
- Internal processing modules (ASR, recommendation engine, RAG)

Data Usage

- Speech-to-text conversion (ASR processing)
- Intent detection and recommendation selection
- Response generation and TTS output
- System performance monitoring and debugging
- Model evaluation and improvement (offline, if allowed)

Data Storage Requirements

- Audio data shall be stored in compressed format

- Text data shall be stored in Unicode (UTF-8) Amharic format
- Conversation data shall be linked using a unique session ID
- Temporary audio data may be deleted after processing if not required

Data Retention Policy

- Raw audio data retained for a limited time
- Transcribed text retained longer for analysis and auditing
- Retention duration configurable by system administrator

Data Privacy and Security

- Speech and text data shall be access-controlled
- Sensitive information shall not be exposed to unauthorized users
- Data shall be encrypted during transmission and storage

Dependencies

- Amharic ASR Module
- Recommendation Engine
- Response Generation and TTS Modules

10.2 Knowledge Base Content

Description

The system shall maintain a structured knowledge base that contains validated agronomy advice and market price information used by the recommendation engine and the RAG component. The knowledge base shall serve as the primary source of factual information to ensure accurate, safe, and reliable responses to farmers' questions.

Content Types

- Agronomy guidelines (planting, fertilizer use, pest control, harvesting)
- Crop-specific advisory information
- Market price data for agricultural products
- Frequently asked questions and standard advisory responses
- Safety rules and constraints for agronomy advice

Content Sources

- Pre-existing datasets provided to the project
- Verified agronomy manuals and extension guidelines
- Market information datasets from trusted sources
- Curated question–answer collections used for training and retrieval

Data Structure

- Content shall be organized by crop type, topic, and region where applicable
- Each knowledge item shall include metadata (source, date, category)
- Content shall be stored in a format suitable for retrieval (documents, chunks, or vectors)

Data Usage

- Used by the recommendation engine to select relevant information
- Retrieved by the RAG component for response generation
- Referenced during admin review and system evaluation

Update and Maintenance

- Knowledge base content shall be updatable by authorized administrators

- Outdated information shall be reviewed and replaced
- Versioning shall be maintained to track content changes

Data Quality Requirements

- Content shall be accurate, validated, and domain-specific
- Conflicting or unverified information shall not be included
- Content shall be reviewed before being made available to the system

Access Control

- Only authorized system components may query the knowledge base
- Only authorized administrators may modify or update content

Dependencies

- Recommendation Engine
- RAG Capabilities
- Admin Dashboard

10.3 Logs and Audit Trails

Description

The system shall maintain logs and audit trails to record important system activities, user interactions, and administrative actions. These records support system monitoring, troubleshooting, security analysis, performance evaluation, and compliance with system governance requirements.

Log Types

- Call session logs (call start, call end, duration)
- Speech processing logs (ASR success or failure events)
- Recommendation and RAG processing logs
- Response delivery logs (TTS generation and playback status)

- Error and exception logs
- Administrative activity logs (content updates, configuration changes)

Audit Trail Content

- Timestamp of each recorded event
- Unique session or request identifier
- Component or service involved
- Action performed (e.g., retrieve data, generate response)
- Result status (success, failure, fallback)

Data Usage

- System debugging and error analysis
- Monitoring system performance and reliability
- Security auditing and misuse detection
- Evaluation of service quality and response accuracy

Storage Requirements

- Logs shall be stored in a structured and searchable format
- Log data shall be protected from unauthorized modification
- Storage shall support high-volume event recording

Retention Policy

- Logs shall be retained for a defined period based on system policy
- Critical audit logs may be retained longer for security review
- Old logs may be archived or deleted based on storage limits

Access Control

- Log access shall be restricted to authorized administrators
- Audit trails shall not be accessible to general users
- All access to audit logs shall itself be logged

Privacy and Security

- Logs shall not store unnecessary personal data
- Sensitive information shall be masked or anonymized
- Log transmission and storage shall be encrypted

Dependencies

- All system microservices
- Admin Dashboard
- Security and Authentication modules

11. Change Management Process

Description

The change management process defines how modifications to system requirements, design, or implementation are requested, reviewed, approved, and documented throughout the project lifecycle. This process ensures that changes are controlled, traceable, and aligned with the project objectives and academic constraints.

Change Request Initiation

- Changes may be requested by project team members, supervisors, or evaluators.
- Each change request shall be documented with a clear description and justification.
- Requests shall identify the affected requirement(s) or system component(s).

Change Evaluation

- The project team shall review each change request.
- The impact on scope, timeline, system quality, and dependencies shall be assessed.
- Changes that significantly affect project objectives may be rejected or deferred.

Approval Process

- Minor changes may be approved internally by the project team.
- Major changes shall require approval from the project supervisor.
- Approved changes shall be assigned a unique change identifier.

Implementation of Changes

- Approved changes shall be implemented in a controlled manner.
- Affected documents (SRS, design diagrams, test cases) shall be updated accordingly.
- Version control shall be used to track modifications.

Change Documentation

- All approved and rejected changes shall be recorded in a change log.
- The change log shall include the request date, decision, and implementation status.
- Historical versions of requirements shall be retained for reference.

Communication of Changes

- Approved changes shall be communicated to all project members.
- Updated documents shall be shared through the project repository.

Dependencies

- Requirements Traceability Matrix
- Project documentation repository
- Supervisor approval workflow

12. Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) is used to ensure that all system requirements are covered, implemented, and tested. It links each requirement to its design, implementation, and test cases.

Requirements Traceability Matrix

Requirement ID	Requirement Description	Source	Design / Module	Implementation	Test Case ID	Status
FR-01	User can register an account	SRS	User Management Module	Registration API	TC-01	Completed
FR-02	User can log in using username and password	SRS	Authentication Module	Login Service	TC-02	Completed
FR-03	System validates user credentials	SRS	Security Module	Validation Logic	TC-03	Completed

FR-04	User can view dashboard after login	SRS	UI Module	Dashboard Page	TC-04	Completed
NFR-01	System responds within 2 seconds	Performance Req.	System Architecture	Backend Optimization	TC-05	Completed
NFR-02	System ensures data security	Security Req.	Security Design	Encryption & Access Control	TC-06	Completed

13. Regulatory and Standards Alignment

The system is designed to comply with relevant regulations, standards, and best practices applicable to voice-based information systems and data handling in Ethiopia.

Area	Standard / Guideline	Alignment Description
Data Privacy	Ethiopian Data Protection Principles	User data is stored securely and accessed only by authorized roles
Information Security	ISO/IEC 27001 (Best Practice)	Encryption, access control, and audit logging are implemented
AI Ethics	Responsible AI Principles	System provides explainable, safe, and non-harmful responses
Telephony	Telecom Service Guidelines	System operates within standard voice call constraints
Accessibility	Inclusive Design Principles	Voice-based interaction supports low-literacy users